19

Lecture 19

5G Core Network Architecture

معماری هستهٔ نسل پنجم

2G — 3G — 4G — **5G** — 6G

COURSE PART
Part IV – 5G system and procedures (Lectures 14, 19–22)

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

DIFFICULTY
Intermediate to advanced

PREREQUISITES
Lectures 9, 12, 13, 14 and 18

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-19-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Draw** the 5G system architecture in reference-point form and **map** each NF to its EPC predecessor.
- **LO2** **Explain** the 5G identities (SUPI, SUCI, 5G-GUTI, GPSI, PEI) and how SUCI protects subscriber privacy.
- **LO3** **Describe** PDU sessions, their types and the session and service continuity (SSC) modes.
- **LO4** **Design** UPF topologies with uplink classifiers and local anchors for edge computing, and **estimate** the latency and backhaul gains.
- **LO5** **Explain** RM/CM states, registration areas and interworking with the EPC.

2 Prerequisite knowledge

You need the EPC architecture (Lectures 9, 12, 13), the 5G overview (Lecture 14) and the service-based mechanics of Lecture 18 (NRF, SBI). This lecture describes *what* the 5G core consists of; Lecture 20 describes the procedures that use it, Lecture 21 slicing and Lecture 22 QoS.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
SUPI / SUCI	شناسهٔ دائمی / پنهان‌شدهٔ مشترک	Permanent identity / its encrypted form sent over the air.
5G-GUTI	شناسهٔ موقت سراسری	Temporary identity allocated by the AMF.
PDU session	نشست PDU	Association between UE and a data network through the 5GC.
DNN	نام شبکهٔ داده	Name of a data network (the 5G equivalent of an APN).
SSC mode	حالت پیوستگی نشست و خدمت	Rules for keeping or changing the session anchor during mobility.
PSA	لنگر نشست PDU	UPF terminating N6 towards the data network.
UL CL / BP	طبقه‌بند فراسو / نقطهٔ انشعاب	UPF functions that split traffic between anchors.
RM / CM	مدیریت ثبت / اتصال	Registration and connection management states.
N26	واسط N26	AMF–MME interface for EPC–5GC interworking.
N3IWF	تابع همکاری غیر 3GPP	Gateway for untrusted non-3GPP access (e.g. Wi-Fi).

3 Introduction and motivation

The 5G core (5GC) is the first mobile core designed from the outset for cloud deployment. It keeps the proven concepts of the EPC – a mobility anchor in the control plane, a session anchor in the user plane, a subscriber database, policy control – but reorganises them: access and mobility are separated from session management, the user plane is fully controlled through PFCP, subscription data are split between front-ends and a repository, and every control-plane function is a service producer (Lecture 18) [7]. The new structure supports features the EPC could not offer natively: access-agnostic registration (3GPP and non-3GPP), multiple user-plane anchors per session for edge computing, Ethernet and unstructured sessions, network slicing and a flow-based QoS model.

4 Fundamental concepts

4.1 Network functions and reference points

Figure 1 shows the 5G system in reference-point representation. Each reference point is realised on the SBI by the services of the producer (e.g. N11 by Nsmf services, N8 by Nudm services),

except N1, N2, N3, N4, N6 and N9.

Table 1. 5GC network functions and their EPC counterparts.

5GC	Main responsibilities	EPC counterpart
AMF	Registration, connection, reachability, mobility; NAS termination and security; SMS	MME (mobility part)
SMF	PDU sessions, IP allocation, UPF selection and control, charging trigger	MME (session part), SGW-C, PGW-C
UPF	Packet routing, forwarding, QoS enforcement, usage reporting, anchoring	SGW-U, PGW-U
AUSF	Authentication server	HSS (AuC part)
UDM / UDR	Credentials processing, subscription data / repository	HSS
PCF	Policy for access, mobility, sessions and UE	PCRF
NSSF	Slice selection	–
NEF	Exposure of capabilities to external applications	SCEF
NRF, SCP, SEPP	Discovery, indirect communication, roaming security	DNS, DRA, DEA

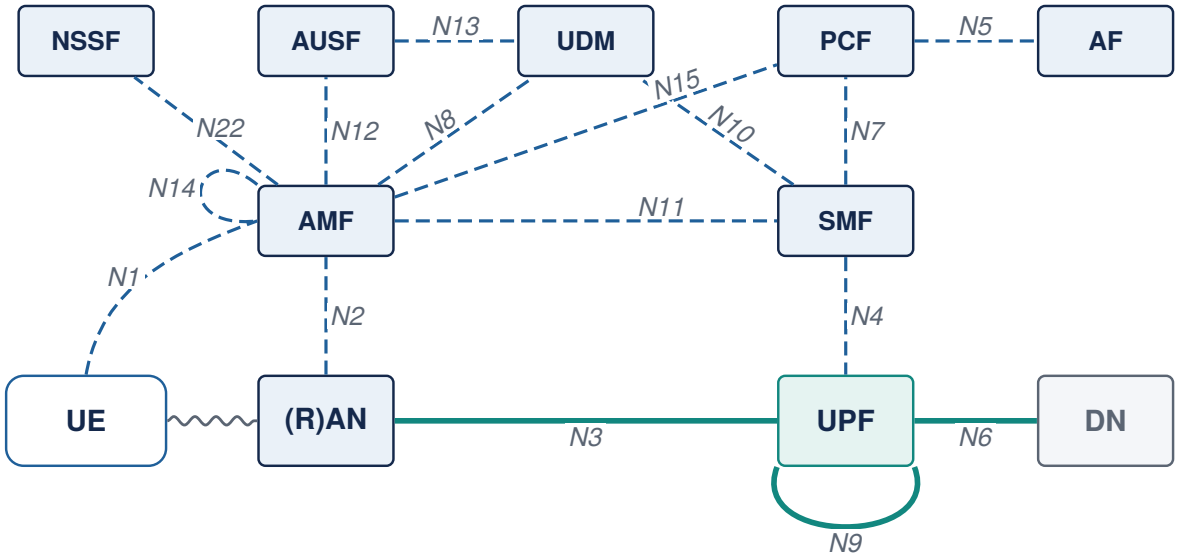


Figure 1. 5G system architecture in reference-point representation (non-roaming, selected NFs) [7].

4.2 Identities

- **SUPI**: the permanent subscription identifier, usually an IMSI (or a network access identifier).
- **SUCI**: the SUPI concealed with the home network's public key; only the routing part is in clear (Figure 2). The UDM's de-concealment function recovers the SUPI [6].
- **5G-GUTI**: temporary identity = GUAMI (PLMN, AMF region, AMF set, AMF pointer) + 5G-TMSI; the 48-bit 5G-S-TMSI is used in paging and RRC.
- **PEI**: permanent equipment identifier (IMEI); **GPSI**: generic public identifier (MSISDN or

external ID) used towards external applications.

Unlike LTE, a 5G UE never sends its permanent identity in clear over the radio interface: even the first registration uses a SUCI, which closes the "IMSI catcher" weakness of earlier generations.

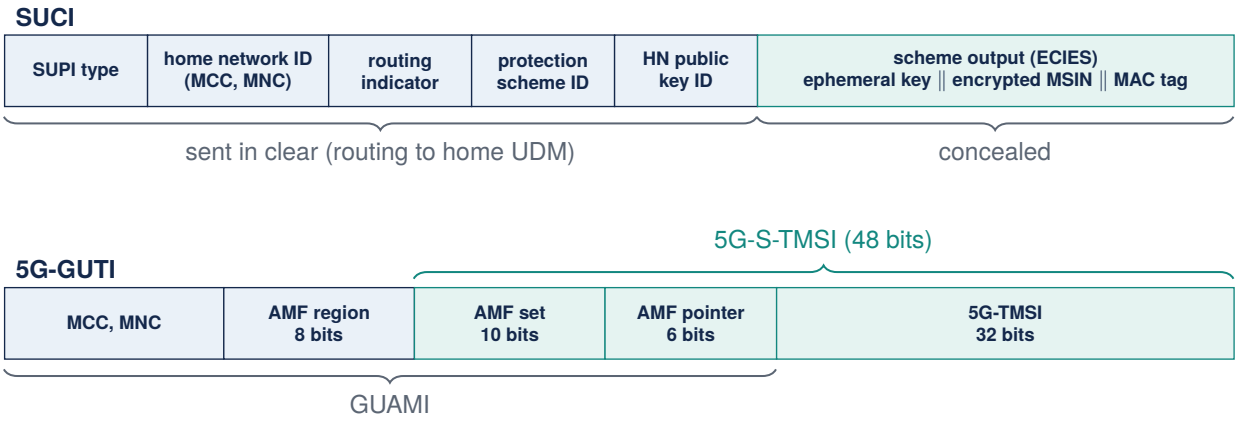


Figure 2. Structure of the SUCI and of the 5G-GUTI.

4.3 PDU sessions

A PDU session connects the UE to a data network identified by a DNN and a slice (S-NSSAI). Session types are IPv4, IPv6, IPv4v6, *Ethernet* (the UPF switches Ethernet frames, e.g. for industrial LANs) and *Unstructured* (opaque data). Within a session, traffic is carried in QoS flows (Lecture 22); a UE may have several sessions to different DNNs or slices at the same time.

4.4 Session and service continuity

The *SSC mode* of a session defines what happens to its anchor when the UE moves [7]:

- **SSC mode 1**: the PSA (and the IP address) is kept for the life of the session – like the EPC's P-GW.
- **SSC mode 2**: the network may release the session and ask the UE to establish a new one with a closer anchor ("break before make").
- **SSC mode 3**: the network establishes a new session with a new anchor before releasing the old one ("make before break"); the application can migrate gracefully.

5 Detailed technical explanation

5.1 The AMF

The AMF terminates N1 (NAS) and N2 (NGAP) [2], [3]. It handles registration, reachability and paging, connection management, mobility and access authentication with the AUSF. Unlike the MME it does not process session signalling: session NAS messages are relayed transparently to the SMF. One AMF serves a UE across 3GPP and non-3GPP access, so registration is access-agnostic. AMFs are organised in *AMF sets*: any AMF of a set can take over a UE's context from

shared storage.

5.2 The SMF and UPF

The SMF manages PDU sessions: it selects UPFs, allocates IP addresses, installs PFCP rules (PDR, FAR, QER, URR, BAR – Lecture 13) over N4, interacts with the PCF for policy and with the CHF for charging. The UPF is the only user-plane function: an *intermediate UPF* (I-UPF) may connect the RAN, an *uplink classifier* (UL CL) or *branching point* (for IPv6 multi-homing) may divert part of the traffic, and one or more *PDU session anchors* (PSAs) terminate N6 (Figure 3).

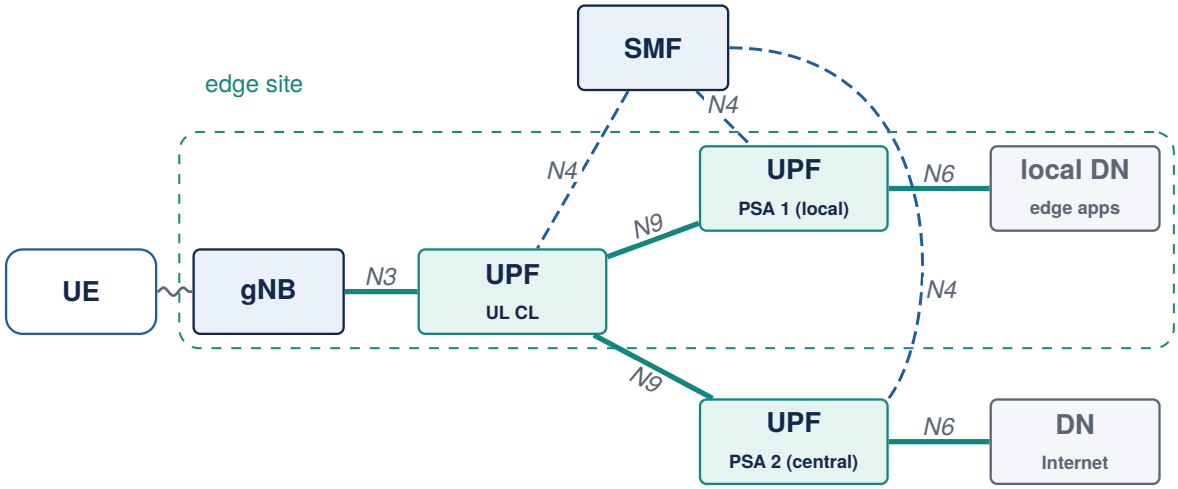


Figure 3. A PDU session with an uplink classifier and two anchors: local edge breakout and central Internet access.

5.3 Edge computing

To serve low-latency applications, the SMF can insert a UL CL and a local PSA at an edge site, so that traffic for edge application addresses is routed locally while the rest goes to the central anchor. Application functions can influence this routing through the NEF or PCF, and TS 23.548 adds discovery of edge application servers [1]. The result is a single PDU session with both local and central breakout – impossible in the EPC without special arrangements.

5.4 States and registration areas

RM states: RM-DEREGISTERED and RM-REGISTERED; CM states: CM-IDLE and CM-CONNECTED. A UE in CM-CONNECTED may additionally be in RRC_INACTIVE: the N2 connection and context stay in place while the RAN pages the UE within a RAN notification area, which saves signalling for frequent small transmissions. The AMF allocates a *registration area* (a TAI list) and performs mobility and periodic registration updates, as the MME did with TAUs.

5.5 Interworking with the EPC

During the long coexistence of 4G and 5G, a UE moves between LTE (EPC) and NR (5GC). With *N26* the MME and AMF exchange contexts directly and IP sessions survive the transition; this requires combined nodes: SMF+PGW-C, UPF+PGW-U, HSS+UDM and PCF+PCRF serve both systems (Figure 4). Without N26 the UE re-registers and the session continuity depends on the

combined anchor. Voice is supported by EPS fallback (a 5G voice call is handed over to LTE for VoLTE) where VoNR is not available.

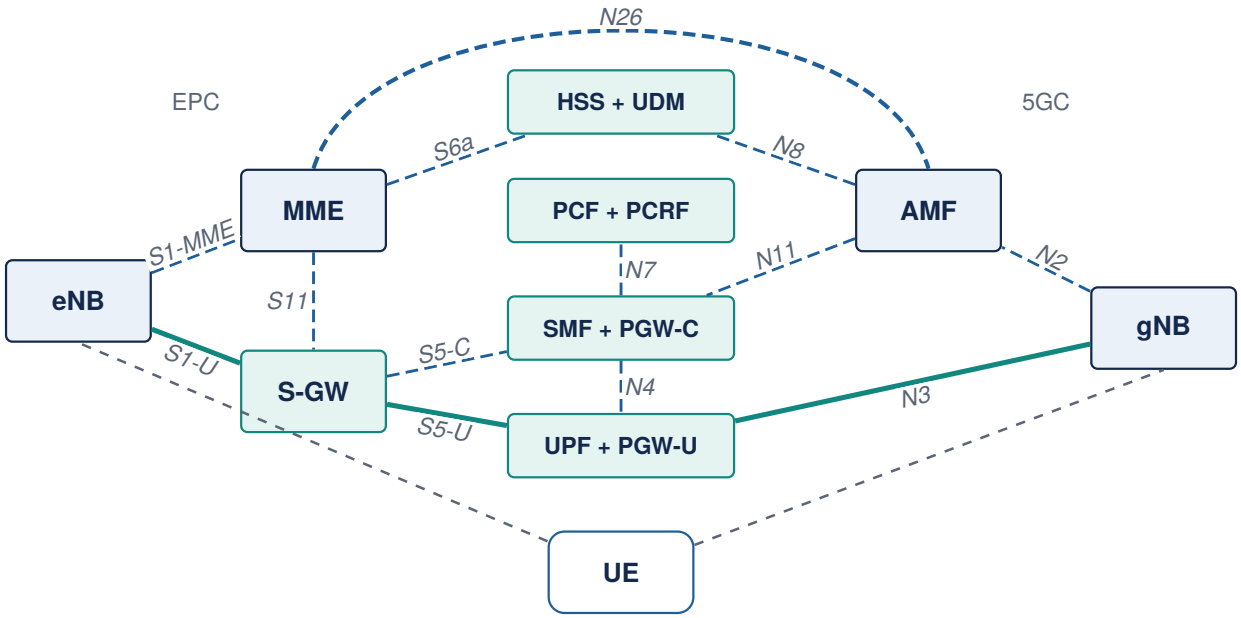


Figure 4. EPC–5GC interworking with N26 and combined nodes.

5.6 Non-3GPP access

Untrusted non-3GPP access (e.g. public Wi-Fi) connects through an N3IWF, which terminates an IPsec tunnel from the UE and presents N2/N3 towards the core; trusted non-3GPP access uses a TNGF. In both cases the UE registers with the same AMF and uses the same NAS and PDU session procedures – the access-agnostic design of the 5GC.

6 Mathematical models

6.1 SUCI size

Model. With ECIES, the scheme output consists of the ephemeral public key (k bytes), the encrypted MSIN (the same length m as the plaintext, BCD encoded) and a MAC tag (t bytes):

$$L_{\text{out}} = k + m + t. \tag{1}$$

Example. A 10-digit MSIN ($m = 5$ bytes): profile A (X25519, $k = 32$, $t = 8$) gives 45 bytes; profile B (P-256 compressed point, $k = 33$) gives 46 bytes [6]. The overhead is small compared with the privacy gained, and the computation is done once per registration attempt that needs a permanent identity.

6.2 Latency and UPF placement

Model. Round-trip latency between UE and an application server behind a PSA at fibre distance D is

$$\mathrm{RTT} \approx \mathrm{RTT}_{\mathrm{radio}} + 2\,\frac{D}{v} + T_{\mathrm{proc}}, \tag{2}$$

with $v \approx 2 \times 10^8$ m/s in fibre (about 5 µs per km).

Example. $D = 500$ km (central data centre) adds 5 ms; $D = 20$ km (edge site) adds 0.2 ms. With a 4 ms radio RTT and 1 ms processing, the RTT falls from 10 ms to 5.2 ms – the edge halves it.

Interpretation. Propagation is only one term; for URLLC targets of a few milliseconds every component – radio numerology, scheduling, transport and processing – must be engineered together.

6.3 Backhaul offload

Model. If a fraction α of a site's traffic T is served locally (UL CL to a local PSA and edge cache or application), the traffic carried towards the central core is

$$T_{\mathrm{bh}} = (1 - \alpha)\,T. \tag{3}$$

Example. With 60 % of video served from an edge cache, a site with 10 Gbit/s busy-hour traffic needs 4 Gbit/s of backhaul towards the central core instead of 10.

6.4 Interruption in SSC modes

Model. When the anchor changes, the service interruption is approximately

$$T_{\mathrm{int}} \approx \begin{cases} 0 & \text{SSC 1 (no anchor change) and SSC 3 (make before break),} \\ T_{\mathrm{rel}} + T_{\mathrm{est}} + T_{\mathrm{app}} & \text{SSC 2 (break before make),} \end{cases} \tag{4}$$

where T_{rel} and T_{est} are the release and establishment times and T_{app} the time for the application to reconnect with the new address.

Example. $T_{\mathrm{rel}} = 50$ ms, $T_{\mathrm{est}} = 150$ ms, $T_{\mathrm{app}} = 300$ ms: half a second without service in SSC 2 – acceptable for a background download, not for a video call, which should use SSC 1 or 3.

7 Network architecture and diagrams

A national 5GC typically has a few central sites hosting AMF sets, SMFs, UDM/UDR, PCF, NRF and the central UPFs, and many edge sites with UPFs (and possibly edge application platforms) close to the RAN. The SMF chooses, per session, which UPFs to chain: for an Internet session a central PSA, for an industrial Ethernet session a local PSA on the factory premises, for a session with edge services a UL CL with both (Figure 3). Slices (Lecture 21) may use dedicated SMFs and UPFs.

8 Step-by-step procedures

Procedure 1 • SUCI calculation and de-concealment

1. The USIM (or ME) holds the home network public key and its identifier, and the protection scheme to use.
2. For each registration that needs a permanent identity, the UE generates an ephemeral key pair, derives a shared secret with the home network public key and encrypts the MSIN; it computes the MAC tag.
3. The UE sends the SUCI (routing part in clear plus scheme output) in the Registration Request.
4. The AMF forwards it via the AUSF to the home UDM, whose SIDF uses the private key to recover the SUPI.

Procedure 2 • Adding a local anchor for edge traffic

1. An edge application is deployed and its address range is configured in (or reported through the NEF/PCF to) the SMF.
2. When the UE is in the edge site's area, the SMF selects a UPF there as UL CL and a local PSA, and installs PFCP rules: traffic matching the edge addresses is forwarded to the local PSA, the rest over N9 to the central PSA.
3. The SMF updates the RAN's N3 tunnel to point to the UL CL (via the AMF and N2).
4. When the UE leaves the area, the SMF removes the UL CL and local PSA; the session continues through the central anchor.

9 Worked examples

Worked example 1 • Mapping EPC to 5GC

Problem. Which 5GC functions take over (a) the MME's tracking-area list management, (b) the MME's bearer set-up via S11, (c) the P-GW's IP address allocation, (d) the PCRF's Gx rules?
Solution. (a) AMF (registration areas); (b) SMF (PDU session management, N4 to the UPF); (c) SMF (allocation) with the UPF as anchor; (d) PCF over N7 (Npcf_SMPolicyControl) to the SMF.

Worked example 2 • Edge latency budget

Problem. An AR application needs an RTT below 10 ms. Radio RTT is 4 ms, processing 1 ms. Up to which fibre distance can the application server be placed?
Solution. $4 + 1 + 2D/v \leq 10 \Rightarrow D \leq 5\text{ ms} \times v/2 = 500\,\text{km}$ of fibre path. Fibre routes are typically 1.5 times the straight-line distance, so about 330 km in a straight line – a regional, not necessarily a cell-site, edge.

Worked example 3 • Choosing an SSC mode

Problem. Choose SSC modes for (a) IMS voice, (b) a vehicle downloading map updates, (c) a cloud game that can migrate between edge servers.
Solution. (a) SSC 1 – the IMS registration is bound to the IP address; (b) SSC 2 – a short interruption is tolerable and a local anchor reduces backhaul; (c) SSC 3 – the application opens a connection via the new anchor before the old one is released.

10 Technology comparison

Table 2. EPC and 5GC architecture compared.

	EPC	5GC
Mobility vs session control	Both in MME	AMF and SMF separated
User plane control	Integrated (CUPS optional)	Always via PFCP (N4)
Session types	IP (non-IP via SCEF)	IP, Ethernet, Unstructured
Anchors per session	One P-GW	One or more PSAs, UL CL/BP
Continuity	Anchor fixed	SSC modes 1, 2, 3
Identity privacy	IMSI may be sent in clear	SUCI (always concealed)
Access	3GPP; non-3GPP via ePDG	Access-agnostic AMF (N3IWF, TNGF)
Signalling	Diameter, GTPv2-C	SBI (HTTP/2), NAS, NGAP, PFCP

Continuity of ideas. Many EPC concepts survive with new names: TAI lists become registration areas, APNs become DNNs, bearers become QoS flows, the HSS becomes UDM/UDR/AUSF. Understanding the EPC therefore remains the best preparation for the 5GC.

11 Key takeaways

Key takeaways

- 5GC separates AMF (access, mobility) from SMF (sessions) and controls UPFs only through PFCP.
- SUPI is never sent in clear; SUCI = routing info + ECIES output ($k + m + t$ bytes).
- PDU sessions (IP, Ethernet, Unstructured) can have several anchors; UL CL and BP enable local breakout.
- SSC modes 1/2/3 trade anchor stability against optimal routing; SSC 2 interrupts service, SSC 3 does not.
- Edge UPFs cut RTT by $2D/v$ and backhaul by a factor $(1 - \alpha)$.
- Interworking with the EPC relies on N26 and combined nodes (SMF+PGW-C, UPF+PGW-U, HSS+UDM, PCF+PCRF).

12 Exercises

Exercise 1 • Level 1 – conceptual • Who does it?

Which NF performs: (a) paging; (b) IP address allocation; (c) slice selection at registration; (d) SUCI de-concealment; (e) exposure of UE location to an authorised external application?

Exercise 2 • Level 1 – conceptual • Ethernet sessions

Give an application of Ethernet PDU sessions and explain why IP sessions would be inconvenient for it.

Exercise 3 • Level 2 – numerical • SUCI size

Compute the scheme-output length for a 9-digit MSIN under profiles A and B. How many bytes does concealment add compared with the null scheme, which sends the MSIN in clear?

Exercise 4 • Level 2 – numerical • Edge offload

A city has 50 edge sites, each with 8 Gbit/s busy-hour traffic. What edge-served fraction is needed to keep total backhaul to the central core below 200 Gbit/s?

Exercise 5 • Level 3 – analytical • Latency design

A factory needs RTT below 5 ms between a robot controller and the UE. Radio RTT is 2 ms with mini-slots and processing 0.5 ms. Where must the PSA and the controller be, and which session type and SSC mode would you choose? What if the radio RTT were 4 ms?

Exercise 6 • Level 3 – analytical • Interworking without N26

Explain what happens to an ongoing IP session when a UE moves from NR to LTE coverage (a) with N26 and combined anchors, (b) without N26 but with combined anchors, (c) without combined anchors. Which case keeps the IP address?

Solutions

Solution to Exercise 1

(a) AMF (and the RAN for RRC_INACTIVE); (b) SMF; (c) AMF with the NSSF; (d) UDM (SIDF); (e) NEF, obtaining the information from the AMF/UDM.

Solution to Exercise 2

Industrial LANs (e.g. time-sensitive networking between controllers and machines) and enterprise Layer-2 VPNs. The devices use Ethernet protocols and MAC addressing, sometimes non-IP; carrying them over IP sessions would need tunnelling or gateways, adding latency and complexity, whereas an Ethernet session lets the UPF switch frames directly.

Solution to Exercise 3

9 digits in BCD occupy 5 bytes (last half-byte padded). Profile A: $32 + 5 + 8 = 45$ bytes; profile B: $33 + 5 + 8 = 46$ bytes. The null scheme carries the 5-byte MSIN, so concealment adds 40 or 41 bytes.

Solution to Exercise 4

Total traffic $50 \times 8 = 400\,\text{Gbit/s}$; $(1 - \alpha) \times 400 \leq 200 \Rightarrow \alpha \geq 0.5$: half of the traffic must be served at the edge.

Solution to Exercise 5

$2 + 0.5 + 2D/v \leq 5 \Rightarrow D \leq 250\,\text{km}$: in principle a regional site, but transport hops, queueing and jitter consume budget, so the PSA and controller are placed on the factory premises (local UPF). An Ethernet session (if the control protocol is Layer-2) with SSC 1 (fixed local anchor). With 4 ms radio RTT only 1 ms remains: $D \leq 100\,\text{km}$, and in practice the same on-premises design, with little margin.

Solution to Exercise 6

(a) The AMF passes the context to the MME over N26; the session is moved to the EPC through the same SMF+PGW-C/UPF+PGW-U: IP address kept, short interruption. (b) The UE attaches/registers in the EPC itself with "handover" indications; the combined anchor can restore the PDN connection with the same address, but with a longer interruption. (c) A new PDN connection with a new IP address: the application must reconnect. Only (a) and (b) keep the IP address.

13 Further reading

TS 23.501 [7] specifies the architecture and TS 23.502 [5] the procedures; TS 23.503 [4] policy; TS 23.548 [1] edge computing; TS 33.501 [6] security including SUCI; TS 24.501 [3] NAS; TS 38.413 [2] NGAP. Rommer et al. [8] give a thorough textbook account of the 5G core.

References

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[8] S. Rommer, P. Hedman, M. Olsson, L. Frid, S. Sultana, and C. Mulligan, *5G Core Networks: Powering Digitalization*. Academic Press, 2020.